AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# T216A
# Cisco Networking (CCCNA) part A

## Mid-term Examination MTA - Makeup
## Fall Semester 2025/2026

## Date: 3/Dec. /2025

| | |
|---|---|
| **Number of Exam Pages:** 4 (including this cover sheet) | **Time Allowed:** 2h |

## Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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**Part 1: Multiple choices questions- 1 mark per question.** **Total: 5 Marks**

1. What is the function of the CRC value that is found in the FCS field of a frame?
   a. To verify the physical address in the frame.
   b. To compute the checksum header for the data field in the frame.
   c. To verify the integrity of the received frame.
   d. To verify the logical address in the frame.
2. What is the characteristic of the LLC sublayer?
   a. It places information in the frame allowing multiple layer 3 protocols to use the same network interface and media.
   b. It defines software processes that provide services to the physical layer.
   c. It provides the logical addressing required that identifies the device.
   d. It provide delimitation of data according to the physical signaling requirements of the medium.
3. What IPv4 header field identifies the upper layer protocol carried in the packet?
   a. Identification
   b. Protocol
   c. Version
   d. Differentiated services
4. What is the auto-MDIX feature?
   a. It enables a switch to dynamically select the forwarding method.
   b. It enables a device to automatically configure the speed of its interface.
   c. It enables a device to automatically configure the duplex setting of a segment.
   d. It enables a device to automatically configure an interface to use a straight-through or a crossover cable.
5. What will happen if the default gateway address is incorrectly configured on a host?
   a. The host cannot communicate with other hosts in the local network.
   b. The host will have to use ARP to determine the correct address of the default gateway.
   c. The switch will not forward packets initiated by the host
   d. The host cannot communicate with hosts in the other networks.

**Part 2: Answer all four questions – 4 marks per question.** **Total: 16 Marks**

1. Answer the following questions:
   a. In the structure of the IOS Command structure (refer to figure A) there is the keyword or argument; what is the difference between the keyword and the argument? (2 marks)

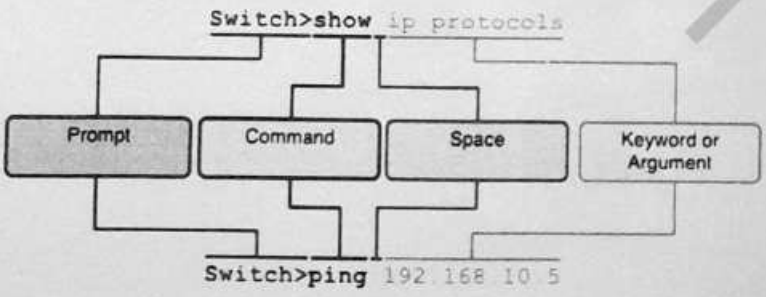



Figure A

SOLUTION



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b. One of the characteristics of the IP protocol is: Best effort, briefly explain this characteristic. (2 marks)

2. Answer the following questions:
   a. There are two types of protocol stacks, the OSI reference model with 7 layers and the TCP/IP protocol suite with 4 layers. (2 marks)

| | | |
|---|---|---|
| OSI model layers: | • Application<br>• Data link layer<br>• Network<br>• Physical | • Presentation<br>• Session<br>• Transport |
| TCP/IP model layers: | • Application<br>• Internet | • Network Access<br>• Transport |

   a1) Which layer(s) of the OSI reference model matches the application layer of the TCP/IP protocol suite. (1 mark)
   a2) Which layer(s) of the OSI reference model matches the Network Access layer of the TCP/IP protocol suite. (1 mark)
   b. Mention one advantage and one disadvantage of Fiber Optic cabling in comparison to cupper cabling (2 marks)

3. Answer the following questions:
   a. Complete the Ethernet frame of figure B by placing the correct field name in its appropriate place (2 marks)
   Field name:
   - Type/Length
   - FCS
   - Source MAC address
   - Data
   - Destination MAC address

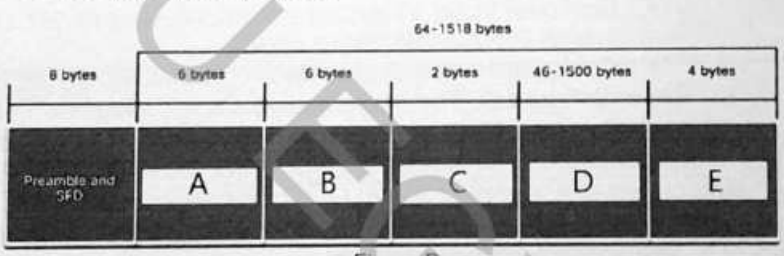


Figure B

   b. What is the purpose of the default gateway in a network. (2 marks)

4. Answer the following questions:
   a. A switch uses forwarding methods to switching data between netowkr ports: list two of these forwarding methods. (2 marks)
   b. When a PC has no entry in the ARP table for an IPv4 address it sent an ARP request message to determine the MAC address of the IPv4 address. What type of destination MAC address will the PC be placing in the header frame/ (2 marks)

**Part 3: Answer all the questions** **Total: 29 Marks**

1. Answer the following questions: (12 marks)
   a) Having the following prefix **199.180.75.0** and subnet mask **255.255.255.240**, determine: (7 marks)
      1. the prefix length (1.5 marks)
      2. the network address (1.5 marks)
      3. the first host address (1.5 marks)
      4. the last host address (1.25 marks)
      5. the broadcast address (1.25 marks)



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b) Convert the decimal number **219**, to binary, then to hexadecimal. Include the steps of calculation. (3 marks)

c) Convert the hexadecimal number DF, to binary, then to decimal. Include the steps of calculation. (2 marks)

**Note:** show the steps of calculation

2. A PC has the following IP address **175.25.99.198** and subnet mask **255.255.255.252**. Determine the network address this PC belongs to. (8 marks)
   a) Convert the dotted decimal IP address **175.25.99.198** to dotted binary; the steps of calculation in details need to be included only for the **first** byte **175** (3 marks)
   b) Convert the dotted decimal subnet mask **255.255.255.252** to dotted binary; (1 marks)
   c) Determine the network address in dotted binary by performing an AND logic between the dotted binary IP address obtained in part a), and the subnet mask obtained in part b). Show the steps of calculation in details. (2 marks)
   d) From the dotted binary network address obtained in part c) determine the network address in dotted decimal. Show the steps of calculation in details only for the **last** byte. (2 marks)
3. Answer the following questions (9 marks)
   a) Write the command(s) needed to configure a switch with the following instructions: (7 marks)
      - Set the name of the router to ***S4-F3*** (1 mark)
      - Configure the following message to be displayed on the screen once a user try to log into the router: ***WARNING: Unauthorized access is prohibited!*** (1 mark)
      - Secure remote Telnet / SSH access with the following password ***7@E3T#*** (2 marks)
      - Configure VLAN 1 interface with the following IP 192.180.0.200 and subnet 255.255.255.224 (2 marks)
      - Configure the default gateway on the switch (192.180.0.193) (1 mark)
   b) Reger to figure C. The figure shows a small switched network and the contents of the MAC address table of the switch. PC4 has sent frame addressed to PC1. What will the switch do with the frame? (2 marks)

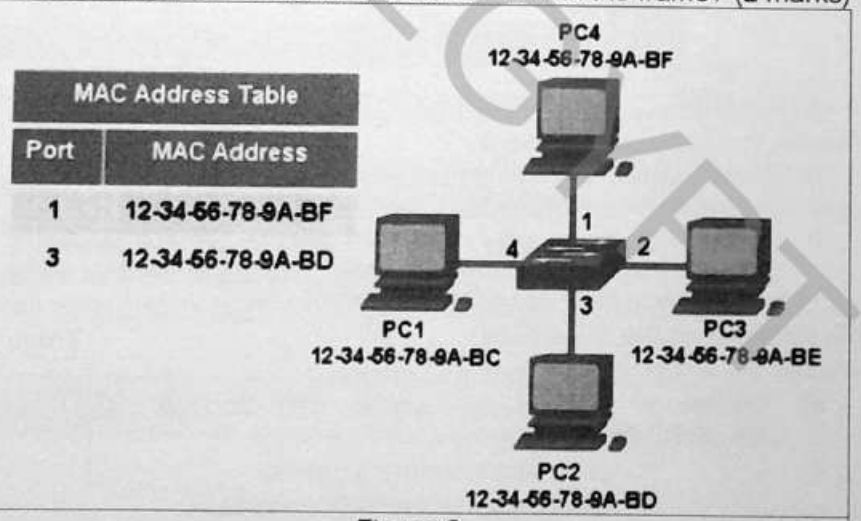



Figure C

**End of questions**



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